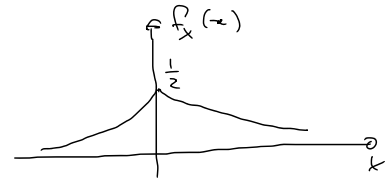
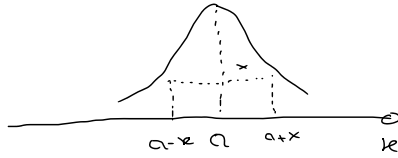


$$f_X(x) = \frac{1}{2} e^{-|x|}$$



$$E[X] = 0$$

$$E[X^2] = \sigma_X^2$$



$$y = x - a \Leftrightarrow x = y + a$$

$$f_X(a+x) = f_X(a-x) \quad \forall x \in \mathbb{R}$$

$$E[X] = a$$

$$= 0 + a \cdot 1 = a$$

$$\int_{-\infty}^{+\infty} x f_X(x) dx = \int_{-\infty}^{+\infty} (y+a) f_X(y+a) dy = \int_{-\infty}^{+\infty} y f_X(y+a) dy + a \int_{-\infty}^{+\infty} f_X(y+a) dy$$

funzione pari
funzione dispari

funzione dispari

$$f_X(x+a) = f_X(x-a)$$

$$\int_{-\infty}^{+\infty} f_X(y+a) dy = \int_{-\infty}^{+\infty} f_X(z) dz = 1$$

$z = y+a$

$$E[X^2] = \int_{-\infty}^{+\infty} x^2 f_X(x) dx = \int_{-\infty}^{+\infty} x^2 \left(\frac{1}{2} e^{-|x|} \right) dx = \frac{1}{2} \int_{-\infty}^0 x^2 e^{-|x|} dx + \frac{1}{2} \int_0^{+\infty} x^2 e^{-|x|} dx$$

$$= 2 \int_0^{+\infty} x^2 f_X(x) dx = 2 \int_0^{+\infty} x^2 \left(\frac{1}{2} e^{-x} \right) dx = \int_0^{+\infty} x^2 e^{-x} dx = \int_0^{+\infty} x^2 e^{-x} dx = 2$$

$f_X(x)$ pari

EXP MONOTONICITÀ

$$f_X(x) = e^{-x} u(x) \rightarrow E[X^2] = \int_{-\infty}^{+\infty} x^2 e^{-x} u(x) dx = \int_0^{+\infty} x^2 e^{-x} dx = 2$$

$1 = \frac{d}{dx} e^{-x}$

$$Y = \frac{1}{\lambda} X \rightarrow E[Y^2] = \frac{1}{\lambda^2} E[X^2]$$

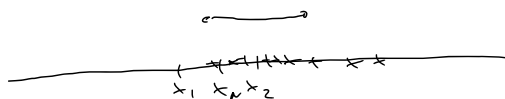
$$\sigma_Y^2 = \frac{1}{\lambda^2} \sigma_X^2$$

$$E[Y^2] = \sigma_Y^2 = \left(\frac{1}{\lambda} \right)^2 \cdot 2 = \frac{2}{\lambda^2}$$

$$f_X(x) = \frac{1}{2} e^{-|x|}$$

$$P[X \in I]$$

$$x_1 \quad x_2 \quad x_n$$



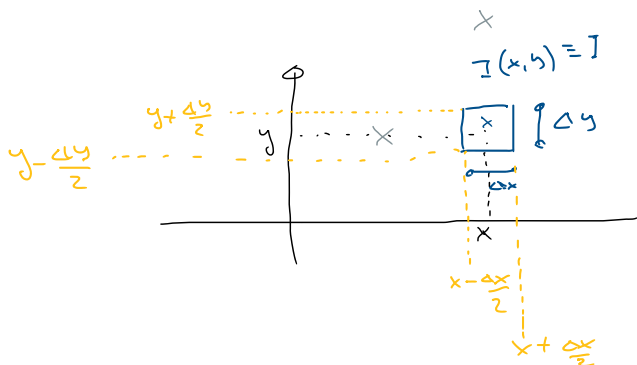
$$P[(X, Y) \in D]$$

$$P[X \in D_m]$$

$$P[X \in D_m]$$

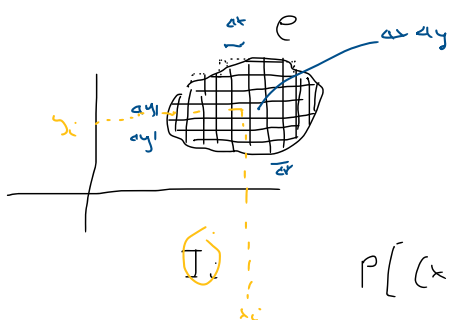


$$P[(X, Y) \in D]$$



$$(X, Y) \in I$$

$$P[(X, Y) \in I] \approx \frac{P[(X, Y) \in I]}{\Delta x \Delta y}$$



$$P[(X, Y) \in I] \approx f_X(x, y) \Delta x \Delta y$$

$\begin{matrix} \Delta x < 1 \\ \Delta y < 1 \end{matrix}$

$$P[(X, Y) \in C] \approx \sum_{i=1}^N P[(X, Y) \in I_i] = \sum_{i=1}^N f_X(x_i, y_i) \Delta x \Delta y$$

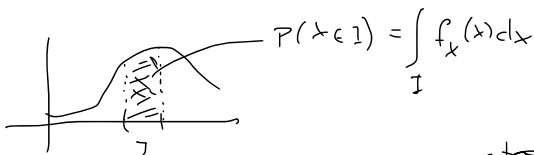
$\begin{matrix} \Delta x < 1 \\ \Delta y < 1 \end{matrix}$

$$P[(X, Y) \in C] = \lim_{\Delta x \rightarrow 0} \lim_{\Delta y \rightarrow 0} \sum_{i=1}^N f_X(x_i, y_i) \Delta x \Delta y = \iint_C f_X(x, y) dx dy$$

$$\int_{-\infty}^{+\infty} \left\{ \int_{-\infty}^{+\infty} f_X(x, y) dx \right\} dy$$

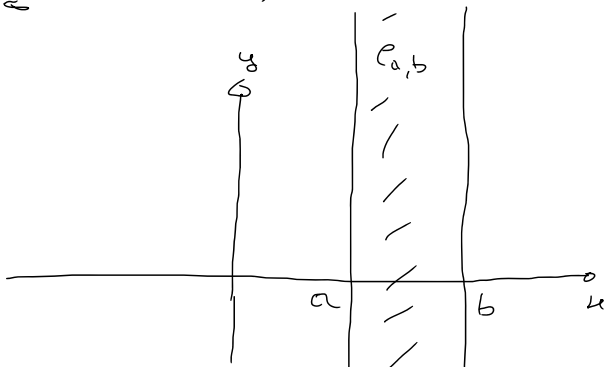


$$\int_C f_X(x, y) dx dy$$



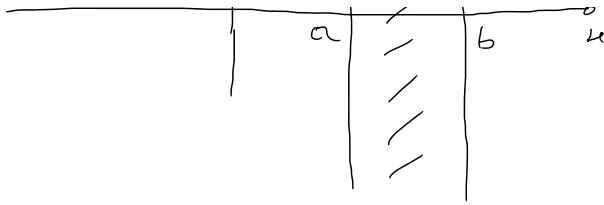
$$P(X \in I) = \int_I f_X(x) dx$$

$$\int_{-\infty}^{+\infty} \left\{ \int_{-\infty}^{+\infty} f_X(x, y) dx \right\} dy = \int_{-\infty}^{+\infty} \left\{ \int_{-\infty}^{+\infty} f_X(x, y) dy \right\} dx$$



$$\int_I f_X(x) dx = 0$$

Assumo

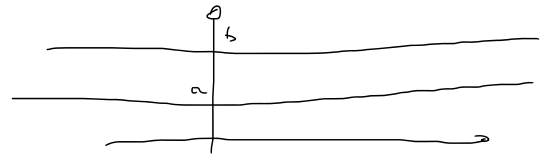


$$(x, y) \in C_{a,b} \Leftrightarrow X \in (a, b)$$

$$\iint_{C_{a,b}} f_{X,Y}(x,y) dx dy = P[(x,y) \in C_{a,b}] = P[X \in (a,b)] = \int_a^b f_X(x) dx$$

$$\int_{-\infty}^{+\infty} \left\{ \int_a^b f_{X,Y}(x,y) dx \right\} dy = \int_a^b \left\{ \int_{-\infty}^{+\infty} f_{X,Y}(x,y) dy \right\} dx$$

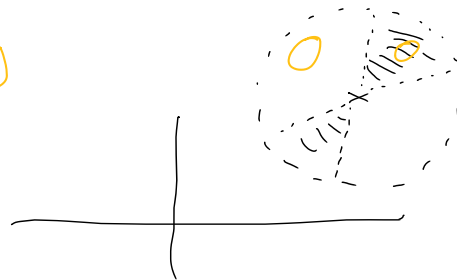
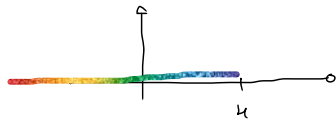
$$f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y) dy$$



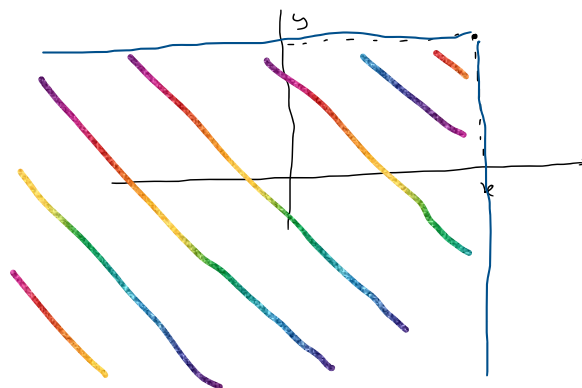
$$\int f_X(x) f_Y(y) dx = f_Y(y)$$

$$\int f_X(x) f_Y(y) dy = f_X(x)$$

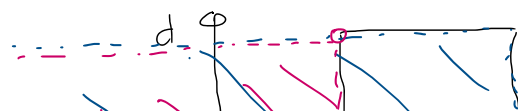
$$F_X(x) = P[X \leq x]$$



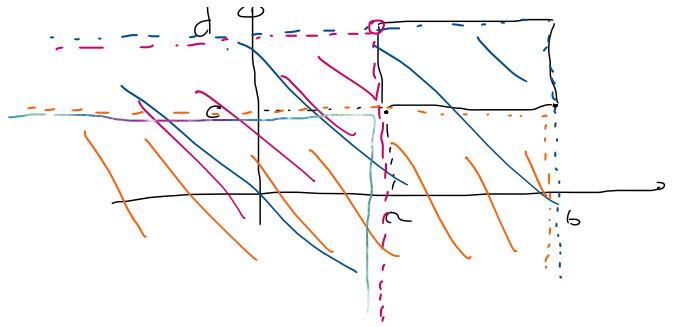
$$F_{X,Y}(x,y) = P[X \leq x, Y \leq y]$$



$$F_X(b) - F_X(a) = \int_a^b f_X(x) dx = P[X \in (a,b)]$$



$$\int_a^b \int_c^d f_{X,Y}(x,y) dx dy = F_{X,Y}(b,d) - F_{X,Y}(a,d) - F_{X,Y}(b,c) + F_{X,Y}(a,c)$$

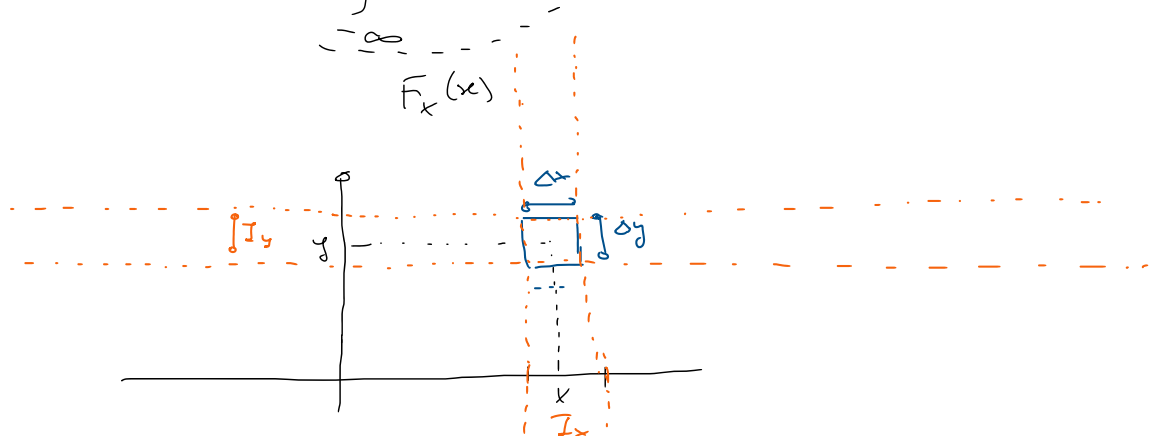


$$P[X \leq x, Y \leq y] = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(x,y) dy dx$$

INDIP.

$$\int_{-\infty}^y f_X(x) f_Y(y) dx = f_X(x) \int_{-\infty}^y f_Y(y) dy = f_X(x) F_Y(y)$$

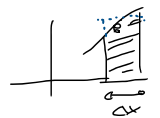
$$\int_{-\infty}^x f_X(x) F_Y(y) dx = F_Y(y) \int_{-\infty}^x f_X(x) dx = F_Y(y) F_X(x)$$



$$P[X \in I_x | Y \in I_y] = P[X \in I_x | Y \in I_y] = \frac{P[A \cap B]}{P[B]} = \frac{P[(X \in I_x) \cap (Y \in I_y)]}{P[Y \in I_y]}$$

$$\int_{I_x} f_X(x) dx = P[X \in I_x]$$

$$f_X(x) = \lim_{\Delta x \rightarrow 0} \frac{P[X \in I_x]}{\Delta x}$$



$$f_{X|Y \in I_y}(x) = \lim_{\Delta x \rightarrow 0} \frac{P[X \in I_x | Y \in I_y]}{\Delta x} = \frac{P[(X \in I_x) \cap (Y \in I_y)]}{P[Y \in I_y]}$$

$$f_{X|Y \in I_y}(x) = \frac{f_{X,Y}(x, y; \Delta x, \Delta y)}{f_Y(y; \Delta y)}$$

$$P(X \in I_x | Y \in I_y) = P(X \in I_x | Y \in I_y)$$

$$x|Y \in I_y$$

$$f_X(x|Y \in I_y) = f_X\left(x \mid y - \frac{\Delta y}{2} < Y < y + \frac{\Delta y}{2}\right)$$

$$f_{X|Y=y}(x) = f_{X|Y}(x|y) = \lim_{\Delta y \rightarrow 0} f_{X|Y \in I_y}(x) = \lim_{\Delta y \rightarrow 0} \lim_{\Delta x \rightarrow 0} \frac{P[(X \in I_x) \wedge (Y \in I_y)]}{\Delta x P[Y \in I_y]}$$

$$= \lim_{\Delta x \rightarrow 0} \lim_{\Delta y \rightarrow 0} \frac{f_{X,Y}(x,y) \Delta x \Delta y}{\Delta x f_Y(y) \Delta y} = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$f_{X,Y}(x,y) = f_Y(y) f_{X|Y}(x|y)$$

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

$$f_{X|Y}(x|y) = f_X(x)$$

$$f_{X,Y}(x,y) = f_X(x) f_{Y|X}(y|x)$$

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y) f_Y(y)}{f_X(x)}$$

$$f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y) dy = \int_{-\infty}^{+\infty} f_Y(y) f_{X|Y}(x|y) dy$$

$$f_{X,Y}(x,y) f_Y(y)$$

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{\int_{-\infty}^{+\infty} f_{X,Y}(x,y) dy}$$

$$P[C_D] = \frac{2}{6} = \frac{1}{3}$$

D_2

D_1, D_3

D_1, D_2

D_1, D_3

D_2, D_1

D_2, D_3

D_3, D_1

D_3, D_2

$$\bar{C}_D = \bar{C}_O$$

$$P[\mathcal{L}_i = (6,6) | C_D] = u^2$$

$$P[X=k|C_D] = \left(\frac{35}{36}\right)^{k-1} \frac{1}{6}$$

$$P[X=k|C_D] = (1-u^2)^{k-1} u^2$$

$$P[X=k|e_3] = \left(\frac{35}{36}\right)^{k-1} \frac{1}{6}$$

$$P[X=k|e_D] = \left(1 - \frac{4}{6}\right)^{k-1} \frac{1}{6}$$

$$P[X=k|\bar{e}_D] = \left(1 - \frac{4}{6}\right)^{k-1} \frac{1}{6}$$

$$(1-p)^{k-1} p$$

$$\sum_{k=1}^{\infty} 1 = 1 \quad \sum_{k=1}^{\infty} k (1-p)^{k-1} p$$

$$\boxed{p = \frac{4}{6}} \\ p = \frac{1}{36}$$